

# **Basic Calculations**

## **User Guide**

**Access Limited Distribution**

**XD Insight**

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## Document Control

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|---------|------------|-----------|-----------------------------|
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# 1. Overview

## 1.1. Purpose of the Guide

The purpose of this document is to provide examples of basic SQL functions within PeopleXD Insight. This includes:

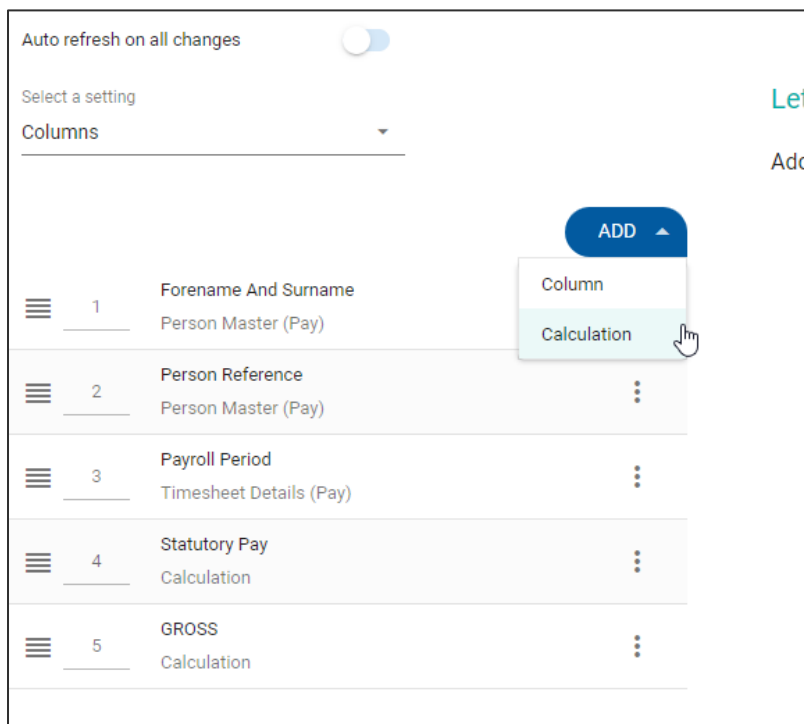
- Writing calculations in Insight
- Data manipulation with functions
- Examples of data that can be returned by calculations

## 1.2. Insight Overview

Insight is PeopleXD's operational reporting tool. Insight utilizes our XD BI layer, which is a collection of database views built from the PeopleXD tables.

Reports are built by choosing the columns from joined views, and applying appropriate filtering to limit the returned data set.

As well as adding data directly from the data columns, it is possible to apply Oracle SQL database functions to manipulate the data. These are referred to as calculation columns. Within the *Edit mode* of a report, to open the new Calculation screen, click **Add**, then **Calculation**:



You can return various types of data with calculations, for example:

- Hardcode Values (e.g. Yes, X, Y etc.)
- Dynamic Values (e.g. sysdate)
- Conditional values (e.g. Age > 40 return A)
- Formatted data from reporting columns (e.g. substring, rounded numbers, conversions etc.)
- Aggregated data, and logical and conditional operations to the data.
  - Additions, subtractions etc.
  - Case Statements, Decodes
  - Counts, Sums

### 1.3. Oracle PL-SQL Functions

Insight runs on an Oracle database. Calculated columns provide you with the ability to call the Oracle SQL functions as part of your report writing.

SQL report writing is a large area, however it is possible to use these functions to format and put some logic in your reports. This requires report writing/BI Developer expertise and experience to apply calculations to achieve advanced reporting requirements.

SQL functions are not unique to Insight, and exist within every database and other software (e.g excel). Using the complex functions requires experience in SQL report writing. Further details are available online when looking for a solution for a reporting calculation, or to enhance your reporting skills.

#### 1.3.1. Document Scope

This document does not cover the full suite of available SQL functions, or more advanced uses or combinations of functions and logic. This resides in the realm of general SQL reporting writing experience, and not the Insight application usage.



**Note:** Only Simple Oracle SQL functions are available through Insight calculated columns. Full procedural SQL (PLSQL) usage is not possible. For example, variables, cursors, IF statements, and other procedural elements are not available in Insight.

## 2. Writing Calculations in Insight

When creating calculated columns in Insight, you can reference any report columns that are in scope based on the columns that exist in your report. The following sections cover the options available once you have added a new calculation to your report.

### 2.1. Manually Create Calculations

To manually create calculations, use the Data Fields tab to search the available data fields. Click on the relevant field to add it to your report, then manually create your calculated column:

| Data Fields      |                        | Calculated Inputs | XD Functions | Calculation Text               |
|------------------|------------------------|-------------------|--------------|--------------------------------|
| name             |                        |                   |              | PRCI_HR_PERSON_MASTER.FORENAME |
| Hr Person Master | Alternative Bank Name  |                   |              |                                |
| Hr Person Master | Alternative Payee Name |                   |              |                                |
| Hr Person Master | Bank Name              |                   |              |                                |
| Hr Person Master | Forename               |                   |              |                                |
| Hr Person Master | Forename And Surname   |                   |              |                                |
| Hr Person Master | Maiden Name            |                   |              |                                |
| Hr Person Master | Middle Name            |                   |              |                                |
| Hr Person Master | Mothers Maiden Name    |                   |              |                                |

The above example, *Forename* is selected from the available columns on the left. This enters the database fieldname *PRCI\_HR\_PERSON\_MASTER.FORENAME* to the Calculation Text on the right.

As an example, from here you could then manually amend it to use it in format function *Upper* which would capitalise the letters:

*upper(PRCI\_HR\_PERSON\_MASTER.FORENAME)*

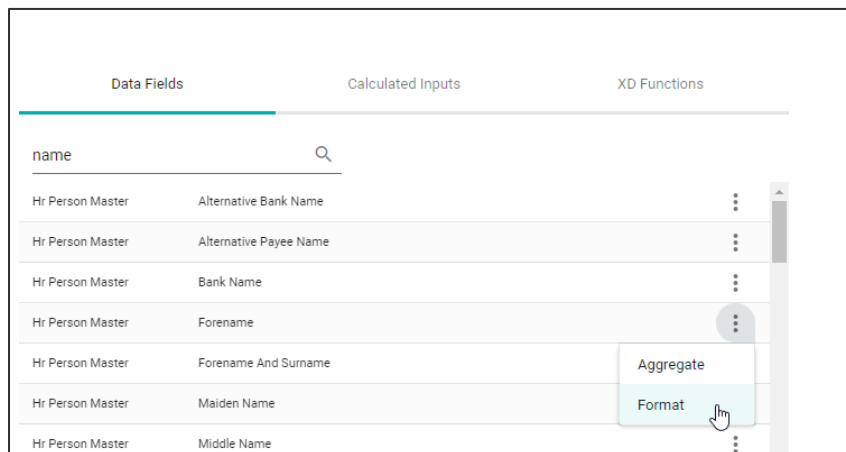
When you save a calculation or function in Insight, the application runs it as a test. An error message will appear if:

1. The syntax is incorrect, or you are referencing columns that don't exist.
  - a. In this scenario, you need to review the syntax and usage of this function.
2. Occasionally, your syntax may be correct, but there may be a conflict. Ensure that the correct columns in your Insight report match the logic being applied in the calculation.

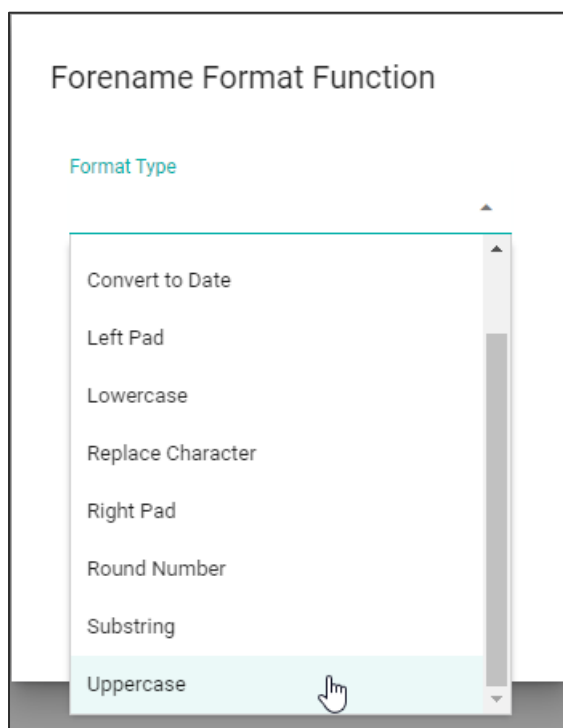
### 2.2. Smart Calculations

You can use the *Smart Calculation* feature if you want to add one of the available aggregate or format functions. To do this, follow the below steps:

1. Click the three dot menu on the relevant data field
2. Click either **Aggregate** or **Format**:



3. Select the required function from the Format Type dropdown list.



For example, the Uppercase function will automatically insert the column as `upper(PRCI_HR_PERSON_MASTER.FORENAME)`.

This can be a good way to start a calculation, even if you need to amend further.

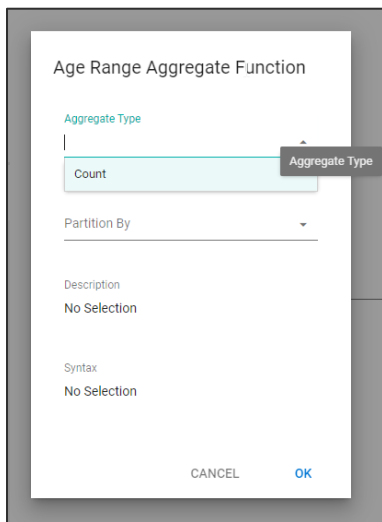
### 2.2.1. Smart Calculation Examples

The sections below illustrate some of the options available with Smart Calculations. These vary depending on the type of column you have selected (i.e. text, number, or date columns).

**Note:** The examples are for illustrative purposes to highlight how Smart Calculations can be used. However, this is not an exhaustive set of examples, further guidance can be found in *PeopleXD Insight: Insight Smart Calculation Builder - User Guide*.

#### Aggregate Examples

In the below example, the only option available for *Age Range* is *Count*. This returns the number of rows in a set of values.



Age Range Aggregate Function

Aggregate Type

Count

Partition By

Description

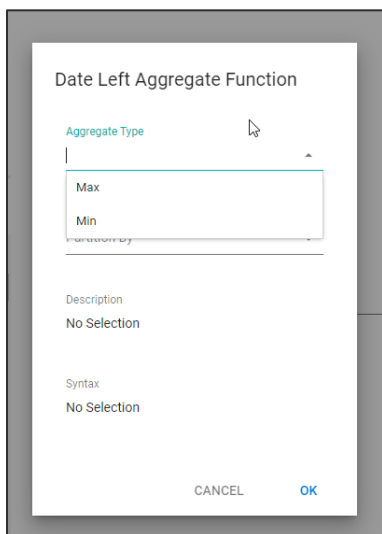
No Selection

Syntax

No Selection

CANCEL OK

In the below example, the options available for *Date Left* are *Max* and *Min*. These return the largest or smallest value in the set of results.



Date Left Aggregate Function

Aggregate Type

Max

Min

Partition By

Description

No Selection

Syntax

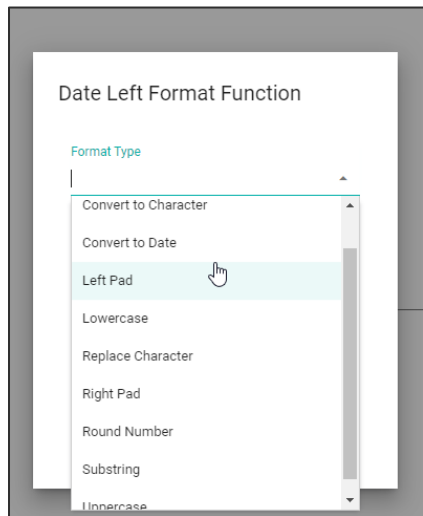
No Selection

CANCEL OK



### Format Examples

The below example shows the formatting options available for *Date Left*. For example, *Left Pad* adds characters to the start of the field.



### 3. Data Manipulation with Functions

This section contains examples of SQL function calculated columns, using the following fields:

- **Person Reference** PRCI\_HR\_PERSON\_MASTER.PERSON\_REFERENCE
- **Forename** PRCI\_HR\_PERSON\_MASTER.FORENAME
- **Surname** PRCI\_HR\_PERSON\_MASTER.SURNAME
- **Date Joined** PRCI\_HR\_PERSON\_MASTER.DATE\_JOINED
- **Home Address1** PRCI\_HR\_PERSON\_MASTER.HOME\_ADDRESS1
- **Home Address2** PRCI\_HR\_PERSON\_MASTER.HOME\_ADDRESS2
- **Home Post Code** PRCI\_HR\_PERSON\_MASTER.HOME\_POST\_CODE
- **Total Salary and Allowance**  
PRCI\_HR\_PERSON\_MASTER.TOTAL\_SALARY\_INCL\_ALLOWANCES
- **Person Department** PRCI\_HR\_PERSON\_MASTER.DEPARTMENT
- **Pay History Value** PACI\_PAY\_HISTORY.VALUE
- **Timesheet Paycode** PACI\_TIMESHEET\_DETAILS.PAY\_CODE
- **Timesheet Value** PACI\_TIMESHEET\_DETAILS.VALUE
- **Person Age** PRCI\_HR\_PERSON\_MASTER.AGE

**Note:** Many of these SQL calculations (e.g. to\_date, to\_char) have a large number of formatting options. This set of examples is not exhaustive, further guidance regarding the functions is freely available online.

#### 3.1. Formatting

The table below gives an example for how each function can be used:

| Function              | Example                                               | Notes                        |
|-----------------------|-------------------------------------------------------|------------------------------|
| Concatenation<br>(  ) | PRCI_HR_PERSON_MASTER.HOME_ADDRESS1                   | Output multiple columns as 1 |
|                       | PRCI_HR_PERSON_MASTER.HOME_ADDRESS2                   |                              |
|                       | PRCI_HR_PERSON_MASTER.HOME_POST_CODE                  |                              |
|                       | Often you will need concatenate some spaces too, i.e. |                              |
|                       | PACI_PERSON_MASTER.HOME_ADDRESS1    ' '               |                              |
|                       | PACI_PERSON_MASTER.HOME_ADDRESS2                      |                              |

| Function                       | Example                                                                                                                                                                                                                                                                                | Notes                                                                                                                                                         |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Capitalise First Letter</b> | initcap(PRCI_HR_PERSON_MASTER.HOME_ADDRESS 1 )                                                                                                                                                                                                                                         | Capitalises the first letter.                                                                                                                                 |
| <b>Convert to Character</b>    | <p>to_char(PRCI_HR_PERSON_MASTER.PERSON_REFERENCE )</p> <p>Other formatting examples:</p> <p>TO_CHAR(sysdate, 'yyyy/mm/dd')</p> <p>Result: '2023/02/09'</p> <p>TO_CHAR(sysdate, 'Month DD, YYYY')</p> <p>Result: 'Feb 09, 2023'</p> <p>TO_CHAR(PACI_PAY_HISTORY.VALUE, '99999.99')</p> | Converts a number or date to character string                                                                                                                 |
| <b>Sysdate</b>                 | Sysdate                                                                                                                                                                                                                                                                                | <p>This function returns the date as of report execution.</p> <p>This is a very useful function often used to supply the current date to other functions.</p> |
| <b>Convert to Date</b>         | <p>to_date(PRCI_HR_PERSON_MASTER.DATE_JOINED )</p> <p>TO_CHAR(SYSDATE,'MONTH')</p> <p><b>Other formatting examples</b></p> <p>TO_DATE('2003/07/09', 'yyyy/mm/dd')</p>                                                                                                                  | Converts a string to a date.                                                                                                                                  |

| Function     | Example                                                                                                                                                                                                 | Notes                                                                                      |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
|              | <p>Result: date value of July 9, 2003</p> <p>TO_DATE('070903', 'MMDDYY')</p> <p>Result: date value of July 9, 2003</p> <p>TO_DATE('20020315', 'yyyymmdd')</p> <p>Result: date value of Mar 15, 2002</p> |                                                                                            |
| Left Pad     | lpad(PCICI_HR_PERSON_MASTER.FORENAME , '2' , 'E' )                                                                                                                                                      | Adds characters to the start of the field. In this example, it will add 'EE'.              |
| Lowercase    | lower(PCICI_HR_PERSON_MASTER.FORENAME )                                                                                                                                                                 | Converts the text to lowercase.                                                            |
| Right Pad    | rpadd(PCICI_HR_PERSON_MASTER.FORENAME , '1' , 'X' )                                                                                                                                                     | Adds characters to the end of the field, in this example it adds 'X'.                      |
| Round Number | round(PCICI_HR_PERSON_MASTER.TOTAL_SALARY_INCL_ALLOWANCES , '0' )                                                                                                                                       | For example, 0 rounds to no decimals places. Entering a 2 will round to 2 decimal places.  |
| Substring    | substr(PCICI_HR_PERSON_MASTER.FORENAME , '1' , '3' )                                                                                                                                                    | Returns part of the string, in this example returns 3 characters starting from position 1. |
| Uppercase    | upper(PCICI_HR_PERSON_MASTER.FORENAME )                                                                                                                                                                 | Converts all text to uppercase.                                                            |

| Function                 | Example                                                                                                                                                              | Notes                                                                                                                                                                                                                                    |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Convert to Number</b> | to_number(PRCI_HR_PERSON_MASTER.TOTAL_SALARY_INCL_ALLOWANCES)                                                                                                        | Converts a string to a number.                                                                                                                                                                                                           |
| <b>Months between</b>    | months_Between(Sysdate, PRCI_HR_PERSON_MASTER.DATE_JOINED)                                                                                                           | Returns the no. of months between two dates. In this case, sysdate (i.e, date of report running and date joined).                                                                                                                        |
| <b>Add Months</b>        | Add_months(PRCI_HR_PERSON_MASTER.DATE_JOINED, 2)                                                                                                                     | In this example, returns a date 2 months after the date joined.                                                                                                                                                                          |
| <b>Average</b>           | AVG(PRCI_HR_PERSON_MASTER.AGE)<br><br>AVG(PRCI_HR_PERSON_MASTER.AGE) OVER (PARTITION BY PRCI_HR_PERSON_MASTER.DEPARTMENT)                                            | In the first example, the average is returned overall. In the second example, it returns average age by department                                                                                                                       |
| <b>Null Value</b>        | NVL(PRCI_HR_PERSON_MASTER.DATE_JOINED,TO_DATE('01-JAN-1900','DD-MM-YYYY'))<br><br>NVL(PRCI_HR_PERSON_MASTER.DATE_JOINED, PRCI_HR_PERSON_MASTER.ORIGINAL_DATE_JOINED) | Null Value returns a default value when a column has no value. In this example, if there is no join date 01-Jan-1900 is returned.<br><br>You could also return the value of another field as the default, as seen in the second example. |

| Function        | Example                                                                                                                                                                   | Notes                                           |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| <b>Truncate</b> | TRUNC(<br>PRCI_HR_PERSON_MASTER.DATE_JOINED,'YEAR')<br><br>TRUNC(<br>PRCI_HR_PERSON_MASTER.DATE_JOINED,'MONTH')<br><br>TRUNC(<br>PRCI_HR_PERSON_MASTER.DATE_JOINED,'DAY') | Returns only the portion of the date specified. |

### 3.2. Aggregation

Aggregation functions perform a calculation on a set of values, and return a single value. For example, count of employees, sum of holiday leave, max no. of absences, sum of payments per pay code, etc.

#### 3.2.1. Count

The table below gives an example for how the count function can be used:

| Function     | Example                                                                                                                                     | Notes                                                                                                                                                              |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Count</b> | count(PRCI_HR_PERSON_MASTER.PERSON_REFERENCE)                                                                                               | Returns the no. of person reference.                                                                                                                               |
|              | count(DISTINCT<br>PRCI_HR_PERSON_MASTER.PERSON_REFERENCE)                                                                                   | The second example is distinct count. For example, if you wanted a unique count of references or view where the reference is used multiple times (e.g. timesheets) |
|              | <b>Count by Department</b><br><br>count(PRCI_HR_PERSON_MASTER.<br>PERSON_REFERENCE) over (partition by<br>PRCI_HR_PERSON_MASTER.DEPARTMENT) | The third example is counting occurrences by department.                                                                                                           |

### 3.2.2. Sum

The table below gives an example for how the sum function can be used:

| Function | Example                                                                                | Notes                                                       |
|----------|----------------------------------------------------------------------------------------|-------------------------------------------------------------|
| Sum      | sum(PACI_PAY_HISTORY.VALUE)                                                            | The first example is summing history values                 |
|          | sum(PACI_PAY_HISTORY.VALUE) over (partition by<br>PACI_PERSON_MASTER.PERSON_REFERENCE) | The second example is summing history values by person ref. |

### 3.2.3. Min/Max

The table below gives an example for how the min/max functions can be used:

| Function | Example                                                                                           | Notes                                                                  |
|----------|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Max/Min  | max(PRCI_HR_PERSON_MASTER.DATE_JOINED)<br>min(PRCI_HR_PERSON_MASTER.DATE_JOINED)                  | Returns the largest or smallest value in the set of results.           |
|          | max(PRCI_HR_PERSON_MASTER.DATE_JOINED)<br>over (partition by<br>PRCI_HR_PERSON_MASTER.DEPARTMENT) | This can be partitioned to a subset of data, in this case, department. |

### 3.2.4. Conditional Logic

Case statements always require a *when*, *then*, and an *end*. The table below gives an example for how case statements can be used:

| Function        | Example                                                                                                                      | Notes                                                                           |
|-----------------|------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Case Statements | CASE WHEN PACI_PAY_HISTORY.PAY_CODE = 530<br>THEN PACI_PAY_HISTORY.VALUE ELSE 0 END                                          | The first two examples output the value for the specific paycodes else output 0 |
|                 | CASE WHEN PACI_PAY_HISTORY.PAY_CODE IN<br>(530, 547, 548, 549, 553, 554, 557, 558) THEN<br>PACI_PAY_HISTORY.VALUE ELSE 0 END |                                                                                 |

| Function | Example                                                                                                                                                               | Notes                                                                                                                                       |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
|          | SUM(CASE WHEN<br>PACI_TIMESHEET_DETAILS.PAY_CODE = '2501'<br>THEN PACI_TIMESHEET_DETAILS.VALUE ELSE 0<br>END)                                                         | Sum pay code 2501,<br>else 0                                                                                                                |
|          | SUM(CASE WHEN<br>PACI_TIMESHEET_DETAILS.PAY_CODE CODE IN<br>(530, 547, 548, 549, 553, 554, 557, 558) THEN<br>PACI_TIMESHEET_DETAILS.VALUE ELSE 0 END)                 | Sum values only for the<br>list of paycodes                                                                                                 |
|          | CASE WHEN PRCI_HR_PERSON_MASTER.AGE >= 20 AND PRCI_HR_PERSON_MASTER.AGE <= 29<br>THEN 'A'<br><br>WHEN PRCI_HR_PERSON_MASTER.AGE >= 30<br>THEN 'B'<br><br>ELSE 'X' END | This is an example of<br>case logic to output an A<br>or B depending on a<br>person's age. Further<br><i>when's</i> could be added<br>here. |

### 3.3. Combination Examples

This section gives calculation snippets of more advanced uses. This is to provide examples of how you can build on the basics, by combining functions in conjunction with logic and arithmetic operations.

 **Note:** This set of examples for SQL functions is not exhaustive, many other advanced options and uses of these functions are available. Further guidance regarding the functions available is freely available online.

Using the complex parameters and options for each function requires a technical skillset and experience in SQL report writing. Further details are available online when looking for a solution for a reporting calculation, or to enhance your reporting skills.



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The table below gives examples of combining functions:

| Example                                                                                                                                                                                         | Notes                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>round((sysdate - PRCI_HR_PERSON_MASTER.ORIGINAL_DATE_JOINED)/365, 2)</b>                                                                                                                     | <p>Adding arithmetic logic to a simple round function.</p> <p>Logic in this example is to get the number of years since original date joined.</p> <p><i>sysdate - date joined</i> gives number of days, dividing by 365 and rounding by two give years e.g., 13.52 years.</p> |
| <b>CASE WHEN CTCI_PERSON_ABSENCE_DETAIL.PAY_CODE = 1123 THEN (CTCI_PERSON_ABSENCE_DETAIL.END_DATE - CTCI_PERSON_ABSENCE_DETAIL.START_DATE) + 1 ELSE 0 end</b>                                   | <p>A conditional case statement using arithmetic logic to get the no. of days (end – start date) for absences records associated with pay code 1123 and 1 day.</p>                                                                                                            |
| <b>MAX(CASE WHEN PACI_PERSON_PENSION_SCHEME.SCHEME_CODE NOT IN(170,171) THEN PACI_PERSON_PENSION_SCHEME.START_DATE END) OVER (PARTITION BY PRCI_HR_PERSON_MASTER.PERSON_REFERENCE)</b>          | <p>When the pension scheme is not 170 or 171, then select the max pension start date.</p>                                                                                                                                                                                     |
| <b>CASE WHEN PRCI_HR_PERSON_MASTER.DATE_JOINED &lt;= SYSDATE+30 THEN 'KEEP' ELSE 'EXCLUDE' END</b>                                                                                              | <p>Output <i>KEEP</i> if the date joined is less than 30 days time, else output <i>EXCLUDE</i> on the report.</p>                                                                                                                                                             |
| <b>CASE WHEN PRCI_HR_PERSON_MASTER.RATE_TYPE_DESC = 'HOURLY' THEN ROUND(PRCI_HR_PERSON_MASTER.RATE_OF_PAY* PRCI_HR_PERSON_MASTER.HOURS*52.143,2) ELSE PRCI_HR_PERSON_MASTER.RATE_OF_PAY END</b> | <p>When the rate of pay is hourly, calculate annual rate, else output just the rate of pay.</p>                                                                                                                                                                               |

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| Example                                                                                                                                                        | Notes                                                                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>NVL(PRCI_HR_PERSON_MASTER.KNOWN_AS,PRCI_HR_PERSON_MASTER.FORENAME)</b>                                                                                      | Defaulting in forename when a <i>known as</i> name doesn't exist.                    |
| <b>CASE WHEN EXCI_TRIP_DETAIL.EXPENSE_CODE = 2100 THEN<br/>EXCI_TRIP_DETAIL.CLAIM_VALUE/6 ELSE EXCI_TRIP_DETAIL.VAT_VALUE END</b>                              | For expense code 2100, divide the value by 6. For other codes, just output the value |
| <b>AVG(PRCI_HR_PERSON_MASTER.AGE) OVER(PARTITION BY<br/>PRCI_HR_PERSON_MASTER.DEPARTMENT)</b>                                                                  | Returns average age for a department.                                                |
| <b>SUM(CTCI_PERSON_ABSENCE_DETAIL.DAYS) over (PARTITION by<br/>CTCI_PERSON_ABSENCE_DETAIL.PERSON_REFERENCE,<br/>CTCI_PERSON_ABSENCE_DETAIL.REASON_CODE)</b>    | Returns the sum of absence days per employee, and absence reason code                |
| <b>max(CTCI_PERSON_ABSENCE_DETAIL.START_DATE) over (PARTITION by<br/>CTCI_PERSON_ABSENCE_DETAIL.PERSON_REFERENCE,<br/>CTCI_PERSON_ABSENCE_DETAIL.PAY_CODE)</b> | Returns newest absence record start date per employee and paycode.                   |